NukesFormers: Unpaired Hyperspectral Image Generation With Nonuniform Domain Alignment

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***Abstract*—The persistent challenge of acquiring precisely co-registered RGB–hyperspectral image (HSI) pairs has significantly impeded the practical deployment of current data-driven hyperspectral image generation (HIG) networks in engineering applications. Gleichzeitig, the ill-posed nature of the aligning constraints, compounded with the complexities of mining cross-domain features, also hinders the advancement of unpaired HIG (UnHIG) tasks. In this article, we conquer these challenges by reformulating the UnHIG through range–null space decomposition (RND), modeling range space feature interaction and null space compensations. Specifically, the introduced contrastive learning effectively aligns the geometric and spectral distributions of unpaired data by building the interaction of range space, considering the consistent feature in the degradation process. Our dual-dimensional contrastive prior module (DCPM) captures mutual information within RGB and HSI domains of a single scene while modeling relationships between internal representations across different scenes, thereby constructing comprehensive cross-domain constraints. Furthermore, the Gabor kernel-based multihead self-attention (G-MSA) adaptively separates high-frequency components, guiding subsequent modules to concentrate on relevant frequency intervals for target objects. Then, we propose a novel nonuniform Kolmogorov–Arnold networks (KANs) (Nukes) to exhaustively excavate null space components (degraded/high-frequency representations) through dual-domain frequency mapping. The proposed method was evaluated on three established datasets: NTIRE 2020 "Clean" track, NTIRE 2022, CAVE, and Grss_dfc_2018. To assess real-world applicability, ratio experiments were conducted by changing the proportion of RGB and HSIs. These experiments demonstrate that our approach achieves state-of-the-art performance in UnHIG.**

***Index Terms*—Dual contrastive learning, hyperspectral image generation (HIG), Kolmogorov–Arnold network (KAN), unpaired data.**

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I. Introduction

HYPERSPECTRAL imaging finds extensive applications in numerous fields, including drug analysis [1], remote sensing detection [2], image fusion [3], and image classification [4], [5]. Nevertheless, due to the inherent limitations of imaging equipment, acquiring an adequate number of hyperspectral images (HSIs) with high quality, compared with RGBs or multispectral images (MSIs), is excessively time-consuming and prohibitively expensive. Furthermore, hardware enhancements tend to occur at a sluggish pace, thereby highlighting the pressing need to rely on software or algorithms to obtain a substantial number of HSIs. Currently, as one of the most efficient technologies for acquiring HSIs, hyperspectral image generation (HIG) enables the generation with both fine spatial and spectral resolution from a single RGB or MSI [6], [7], [8], [9], which is modeled to learn inverse mapping from HSIs.

Similar to other tasks, the data-driven learning strategy in HIG methods has recently witnessed a significant improvement in reconstruction performance [10], [11], as depicted in Fig. 1(a). Nonetheless, the reliability of fully supervised HIG is often reliant on the amount and diversity of paired and registered RGB–HSI or MSI–HSI training sets, which are challenging to obtain in practical work yet.

To tackle these challenges, numerous studies have proposed training HIG models exploiting unsupervised [12], [13], and unpaired methods [14], thereby mitigating the dependency on high-quality paired data with well co-registration. Among these approaches, unpaired HIG (UnHIG) has garnered increasing attention due to the prevalence of nonregistered HSIs in practical applications [14]. However, UnHIG is constrained by the absence of natural cross-domain interactions, leading to a serious nonuniform status between the corresponding information in dual domains. Additionally, the lack of human-annotated hints renders the valuable hyperspectral priors largely inaccessible during the training phase. As a result, critical degraded features, predominantly comprising high-frequency components, are either discarded or overwhelmed in existing approaches. Unfortunately, the challenges still seriously block the performance of UnHIG at present.

Drawing inspiration from the advances of range–null space decomposition (RND) methodologies in various tasks [15], [16], we explore the feasibility of leveraging RND to address the challenges of UnHIG. Specifically, we aim to decompose the inverse high-dimensional mapping into range space interactions encompassing consistent features, and null space

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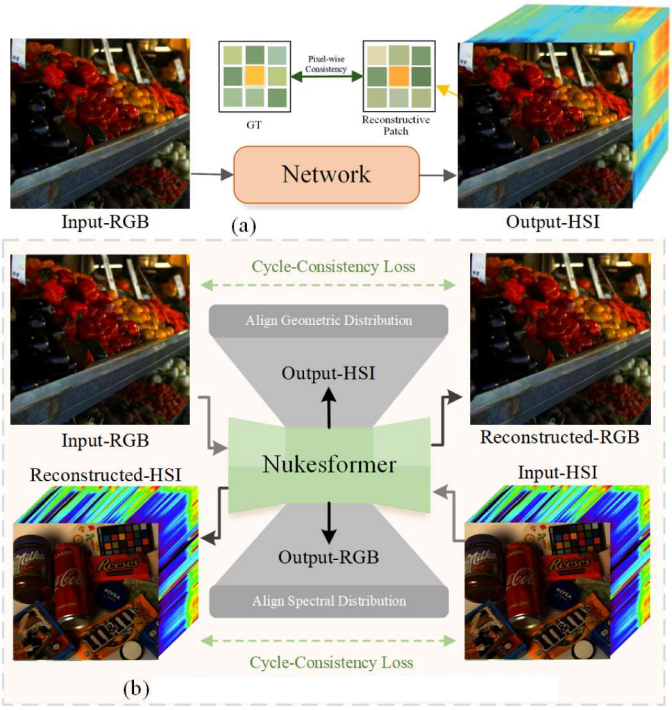


Fig. 1. (a) Conventional framework of paired HIG, which utilizes ground truth (GT) to build direct pixel-wise consistency. (b) Our proposed NukesFormer for UnHIG without GT, which establishes indirect constraint with cycle consistency and DCPM.

compensations targeting degraded and high-frequency components. Within this framework, range space interactions function as an indirect spectral and geometric constraint by encapsulating consistent or inconsistent information inherent to intradomain and intrascene relationships. On the other hand, the null space compensation focuses on mining weak and attenuated high-frequency components of the null space, thereby enhancing the overall reconstruction process.

In this article, we propose a novel and lightweight framework for UnHIG by integrating cross-domain interactions and leveraging spectral characteristic information. Concretely, the unpaired data is systematically decomposed into continuous components termed the range space and attenuated components referred to as the null space, via RND methodology. Subsequently, the framework employs a cycle-based architecture to capture intradomain attributes while utilizing dual-dimensional contrastive learning to aggregate continuous elements within the range space effectively. Moreover, building on Kolmogorov–Arnold networks (KANs) [17], [18], a dedicated nonuniform matrix object-aware mechanism dynamically shifts the B-spline function to enhance null space compensation, facilitating high-dimensional fitting with robust guidance. Finally, extensive experimental evaluations demonstrate that the proposed method achieves state-of-the-art performance across various benchmarks. The key contributions of this work are summarized as follows.

1) We introduce a novel contrastive priori framework for UnHIG, which decomposes the generation process into null space mining and range space alignment, enabling superior high-frequency compensation and effective cross-domain interaction.
2) To adaptively extract high-frequency components obscured within the null space, we propose a nonuniform KANs (NukesFormers), which generates flexible and adaptive rational bases that precisely capture vulnerable yet crucial characteristic bands.
3) A dual-branch contrastive structure is further proposed that independently calibrates both geometric and spectral distributions to facilitate effective cross-domain interaction in range space, leveraging HSIs and RGB or MSIs from different scenes to build reliable constraints.
4) We conducted extensive experiments, encompassing comparative, ablation, and ratio analysis, benchmarking against existing UnHIG frameworks and state-of-the-art models, in which the NukesFormers achieves the best performance across both quantitative metrics and visual comparisons.

The remainder of this article is structured as follows. Section II delineates the existing work on HIG, the development of unpaired methods in computer vision, and the related works about the Transformer with the KAN. Subsequently, Section III details the proposed methods. The experimental results for the comparison, ablation, and ratio are presented in Section IV. Lastly, the conclusion of this article is provided in Section V.

II. Related Works

A. HSI Generation

Recent learning-based HIG models have demonstrated substantial progress in robustness and performance. They are regarded as having a crucial positive impact on existing remote sensing analysis [19], just like the existing analytical methods. These approaches typically leverage large-scale registered paired HSI–RGB (or his–MSI) datasets and employ comprehensive pixel-level constraints, enabling rapid convergence and superior results across diverse scenarios. However, practical deployment is constrained by the frequent scarcity of intact, reliably registered training data, often limited to incompletely registered and limited-scale datasets. In challenging scenarios, only unpaired data may be available, rendering existing direct constraint paradigms inapplicable. From the data-oriented perspective, HIG is categorized into three distinct categories: supervised [7], [8], [20], prior-based unsupervised [12], and unpaired methods.

Representing the most stringent data requirements and the most direct method of constraint construction, the supervised HIG utilizes many paired RGB–HSIs to construct the point-to-point constraint. Zhao et al. [6] proposed that the HRNet employs the residual dense block to obtain a larger receptive field and deploys the PixelShuffle to achieve feature fusion between layers. Consequently, a self-supervised HIG framework was proposed [12], which uses SRF and pretrained semantic segmentation networks to construct point-level constraints, resulting in impressive results. To capture global spectral dependencies, MST++ [8] integrates Transformer architectures into HIG. Addressing the parameter complexity

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of existing models, SPECAT [9] implements cumulative-attention blocks to achieve a lightweight framework. Si et al. [21] reformulated perceptual modeling constraints through discrete wavelet transforms and conditional diffusion models.

The aforementioned approaches neglect the vast collection of existing HSIs that have been captured and stored due to the absence of essential prior knowledge, which implies that unpaired methods potentially reactivate the overlooked data to satisfy the requirements of industrial deployment.

B. Unpaired Methods in Computer Vision

The paradox between a vast amount of redundant unpaired data and the challenge of constructing appropriate constraints has consistently impeded the progress of computer vision, leading to the wastage of collected data [22], [23]. Initially, a Bayesian framework was constructed to establish the connection between two domains [24], utilizing Markov random fields in image-to-image translation.

Additionally, the cycle-based [25] approaches employ paired generators and discriminators to simulate domain information extraction and introduce cycle consistency to achieve intuitive constraints [26]. In deraining, a contrastive prior is utilized to construct positive and negative sample pairs to extract and remove rain signals [23]. By utilizing contrastive learning methods to address patch information at the same position in images and retrieve information from distant parts, effective constraints are indirectly achieved [27]. Ho et al. [28] introduced dense normalization (DN) to effectively mitigate tiling artifacts while preserving local color and hue contrasts through pixel-level statistical moment estimation for unpaired image-to-image translation. SurgicaL-CD [29] integrates diffusion into computer-assisted surgery to generate realistic surgical imagery from simulations using unpaired datasets. For multimodal alignment, Huang et al. [30] proposed multimodal aligned conceptual knowledge, constructing bidirectional similarity pooling across visual and textual domains to substantially enhance zero-shot and cross-dataset image–text matching performance. HEDehazeNet [31] implements a dedicated haze generation enhancement module within the CycleGAN framework, achieving state-of-the-art results with dynamically complex and diverse training samples.

In a similar vein, HIG tasks also strive to leverage ample unmatched HSI and RGB data [12]. This underscores the critical need to develop a model capable of achieving stable training and robust cross-scene interaction while relying solely on unpaired data.

C. Transformer With the KAN

KANs leverage B-spline basis functions to precisely modulate the learned function curves, enabling superior approximation of complex data distributions with reduced parameterization compared to conventional convolutional or linear layers. By eschewing nonlinear activation functions, KANs demonstrate enhanced robustness in feature learning. The compositional nature of intrinsically simple basis functions yields inherently more interpretable representations. Critically, given the substantial parameter inefficiency of conventional feed-forward networks (FFNs) and multilayer perceptrons (MLPs) relative to KANs, KANs offer compelling lightweight and generalizable alternatives within contemporary Transformer-based architectures [32], [33], [34].

Yang and Wang's [34] Kolmogorov–Arnold Transformer (KAT) integrates rational activation and Group KAN for GPU-friendly execution and reduced computational complexity. Through optimization of GR-KAN components, KAT enhances the scalability of KAN-with-Transformer for large-scale applications. TKAT [33] employs recurrent KAN layers to minimize memory overhead while capturing temporal features.

In HSI generation tasks, the implicit/explicit RGB–HSI mapping aligns structurally with KAN's B-spline formulation, suggesting a natural synergy. Integrating these paradigms is expected to improve both interpretability and performance.

III. Method

In this section, we first present the MetaFormer [35] architecture equipped with nonuniform KANs (NukesFormers) tailored for the UnHIG. In NukesFormers, the "degradation" phase [Fig. 2(a)] deploys the dynamic Gabor Transformer to mine high-frequency components and the cascading nonuniform KANs embedded multihead self-attention (Nuk-MSA) to compensate for the attenuated null space in the RGB domain. Then, we introduce the dual-dimensional contrastive prior module (DCPM), designed to build a reliable constraint with range space.

A. NukesFormers for UnHIG

Unlike paired HIG, we employ an acquired HSI denoted $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$ along with an RGB of a different scene, represented as $\hat{\mathbf{Y}} \in \mathbb{R}^{3\times H\times W}$, in UnHIG. Obviously, UnHIG cannot directly leverage GT to extract latent feature representations of the HSI domain, but a large amount of unpaired RGB–HSIs still bury a unified degradation-reconstruction process as below:

$$\mathbf{Y} = \mathbf{D}\mathbf{X} + \mathbf{N}_c \quad \hat{\mathbf{Y}} = \mathbf{D}\hat{\mathbf{X}} + \mathbf{N}_c \tag{1}$$

where $\mathbf{D} \in \mathbb{R}^{3\times C}$ and $\mathbf{N}_c \in \mathbb{R}^{3\times H\times W}$ are depicted as the SRF and camera noise, respectively. Furthermore, $\mathbf{Y}$ means the RGB in the same scene as $\mathbf{X}$.

However, the mere construction of a cycle-based method has been fully explored [25], and significant mode collapse has been observed. The UnHIG urgently requires modeling an implicit relationship between the two modalities across scenes to link their training processes. Naturally, based on further derivation of the degradation-reconstruction process, the RND methodology is a feasible solution.

Omitting $\mathbf{N}_c$ considerations, the dual direction mapping process can be portrayed using the inverse operation (pseudo-inverse) of $\mathbf{D}$ as $\mathbf{D}^\dagger \in \mathbb{R}^{C\times 3}$, as illustrated in the RND methodology [36]

$$\begin{aligned}\mathbf{X} &\equiv \mathbf{D}^\dagger\mathbf{D}\mathbf{X} + \left(\mathbf{I} - \mathbf{D}^\dagger\mathbf{D}\right)\mathbf{X}\\ \hat{\mathbf{Y}} &\equiv \mathbf{D}\mathbf{D}^\dagger\hat{\mathbf{Y}} + \left(\mathbf{I} - \mathbf{D}\mathbf{D}^\dagger\right)\hat{\mathbf{Y}}\end{aligned} \tag{2}$$

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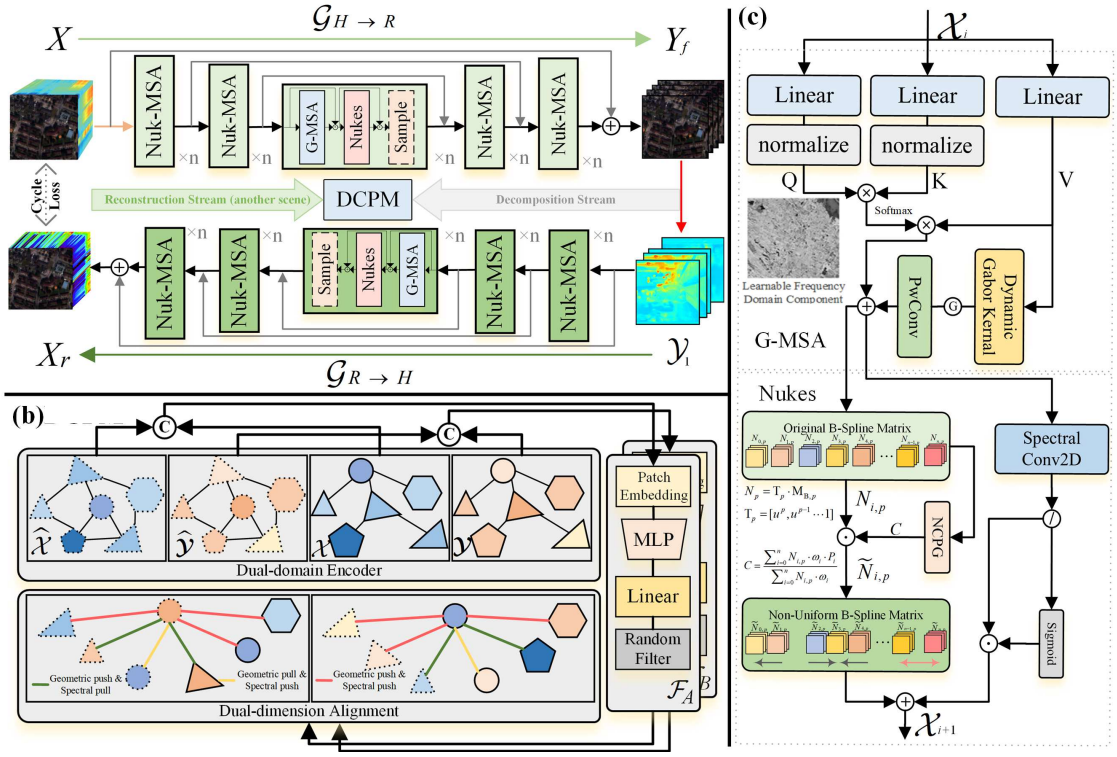


Fig. 2. Overall framework of the proposed NukesFormer of decomposition stream, where the reconstruction stream in (a) derives from another NukesFormer, leveraging shared parameters. (b) DCPM utilizes the dual-dimensional contrastive before pull (→←), similar geometric and spectral distributions, and push (⟷) different components. (c) Nonuniform B-spline matrix and G-MSA are dedicated to capturing multifrequency information from the spectral dimension.

which decomposes the HSIs into range space $\mathbf{D}^{\dagger}\mathbf{D}\mathbf{X}$ and null space $(\mathbf{I}-\mathbf{D}^{\dagger}\mathbf{D})\mathbf{X}$.

RND transforms "finding the connection between two modalities across scenes" into "decomposing the degradation process into the alignment and restoration of two components." For the UnHIG, these two components can be naturally interpreted as spatial and spectral information, since the degradation process is essentially sampling along the spectral dimension. Based on the above analysis, we can rewrite the UnHIG with high interpretability, which is the alignment of the range space and the exploration of the null space. Finally, by combining the deeply aligned range space features and the decompressed null space features, highly interpretable and reliable unpaired HSI generation is achieved.

Meanwhile, the cycle-based framework can be built by degradation-reconstruction operators $\mathbf{D}^{\dagger}\mathbf{D}(\cdot)$ as long as $\mathbf{I}-\mathbf{D}^{\dagger}\mathbf{D}=\mathbf{0}$, which implies that $\mathbf{D}$ must be orthogonal (this, however, is infeasible since $C>3$). As a consequence, we redefine the equation as $\mathbf{min}\|\mathbf{X}-\mathbf{D}^{\dagger}\mathbf{D}\,\mathbf{X}\|_2^2, \mathbf{s.t.}, \|(\mathbf{I}-\mathbf{D}^{\dagger}\mathbf{D})\mathbf{X}\|_2^2=\epsilon_0$, ϵ_0 represents a value infinitesimally close to zero. Obviously, the above constraints can be simplified to $\mathbf{min}\|\mathbf{D}^{\dagger}\mathbf{D}\,\mathbf{X}-\mathbf{X}\|_2^2$. Then the "reconstruction" phase possesses a similar consequent, which establishes the cycle-based constraint as follows:

$$\mathcal{L}_{\text{cyc}}\left(\mathbf{X},\hat{\mathbf{Y}}\right)=\mathbb{E}_{r\sim R}\left[\|\mathcal{G}_{R\to H\to R}\left(\mathbf{X}\right)-\mathbf{X}\|_2^2\right]+\mathbb{E}_{h\sim H}\left[\left\|\mathcal{G}_{H\to R\to H}\left(\hat{\mathbf{Y}}\right)-\hat{\mathbf{Y}}\right\|_2^2\right] \tag{3}$$

where $\mathcal{G}_{R\to H\to R}$ and $\mathcal{G}_{H\to R\to H}$ represent the joint models of reconstruction-degradation ($\mathcal{G}_{R\to H}$-$\mathcal{G}_{H\to R}$) and degradation-reconstruction operations with NukesFormers, respectively, as depicted in Fig. 2(a).

B. Nonuniform KANs Embedded MSA

To obtain the consistent feature of the RGB–HSI domain buried in unpaired data, we introduce the NukesFormer, accurately capturing geometric distributions and essential high-frequency information in null space. As depicted in Fig. 2(a), spectral degradation $\mathcal{G}_{H\to R}$ and reconstruction $\mathcal{G}_{R\to H}$ are individually modeled using paired NukesFormers, which are then integrated into dual domains through a shared parameter mechanism

$$\begin{aligned}\mathbf{Y}_f&=\mathcal{G}_{H\to R}\left(\mathbf{X}|\theta_{H\to R}\right)\quad \mathbf{X}_r=\mathcal{G}_{R\to H}\left(\mathbf{Y}_f|\theta_{R\to H}\right)\\ \hat{\mathbf{X}}_f&=\mathcal{G}_{R\to H}\left(\hat{\mathbf{Y}}|\theta_{R\to H}\right)\quad \hat{\mathbf{Y}}_r=\mathcal{G}_{H\to R}\left(\hat{\mathbf{X}}_f|\theta_{H\to R}\right)\end{aligned} \tag{4}$$

in which $\theta_{R\to H}$ and $\theta_{H\to R}$ represent the learnable parameter. In the "degradation" phase, acquired HSIs $\mathbf{X}$ undergo a modeled degradation process to produce the corresponding fake-RGB $\mathbf{Y}_f$. Subsequently, a NukesFormer simulates the reconstruction process, converting the fake RGB back to a recovery-HSI $\mathbf{X}_r$. In the "reconstruction" phase, the RGB of another scene experiences a similar process.

1) Nuk-MSA: However, one can anticipate that $\mathcal{G}_{H\to R}$ might yield high-frequency components of the HSI domain that are overly degraded, due to the abstraction of intermediate features as the network deepens. Initially, we sample the RGB or HSI to obtain the original feature

$$\mathcal{X}_1=\text{Conv}_{1\times 1}\left(\mathbf{X}\right)\quad \mathcal{Y}_1=\text{Conv}_{1\times 1}\left(\mathbf{Y}_f\right). \tag{5}$$

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Subsequently, these features undergo a progressive refinement process via a series of cascaded Nuk-MSA

$$\begin{aligned}\mathcal{X}_{i+1} &= \mathbf{M}_{\mathrm{DGK}}\left(\mathcal{X}_j \middle| \theta_{en,i+1}^{H\to R}\right) \quad \mathcal{Y}_{i+1} = \mathbf{M}_{DGK}\left(\mathcal{Y}_i \middle| \theta_{en,i+1}^{R\to H}\right)\\ \mathcal{X}_{j+1} &= \mathbf{M}_{\mathrm{DGK}}\left(\mathrm{Map}\left(\mathrm{Concat}\left(\mathcal{X}_j, \mathcal{X}_i\right)\right) \middle| \theta_{de,j+1}^{H\to R}\right)\\ \mathcal{Y}_{j+1} &= \mathbf{M}_{\mathrm{DGK}}\left(\mathrm{Map}\left(\mathrm{Concat}\left(\mathcal{Y}_j, \mathcal{Y}_i\right)\right) \middle| \theta_{de,j+1}^{R\to H}\right). \end{aligned} \tag{6}$$

$\mathcal{X}_i, \mathcal{Y}_i$ are the intermediate features of every encoder in NukesFormer, and $\mathcal{X}_j, \mathcal{Y}_j$ represent the features of the corresponding level in the decoder

$$\mathbf{Y}_f = \mathcal{X}_F + \mathcal{X}_1 \quad \mathbf{X}_r = \mathcal{Y}_F + \mathcal{Y}_1. \tag{7}$$

Finally, a skip connect is utilized to gain $\mathbf{Y}_f$ and $\mathbf{X}_r$.

2) Gabor Kernel-Based Multihead Self-Attention: In contrast to frequency-domain transformations that rely on FFT or similar techniques [37], [38], [39], [40], the utilization of Gabor kernels enables the model to focus more on local and needed frequency intervals, which effectively mitigates the issue of high-frequency information being overshadowed.

Specifically, we employ the spectral multihead self-attention (MSA) to generate the requisite query $\mathbf{Q} \in \mathbb{R}^{HW\times C}$, key $\mathbf{K}^T \in \mathbb{R}^{C\times HW}$, and value $\mathbf{V}$ through linear operations, thus obtaining global spectral features

$$\begin{aligned}\mathbf{Q}, \mathbf{K}, \mathbf{V} &= \mathrm{Linear}\left(\mathcal{X}_i | \theta_Q, \theta_K, \theta_V\right)\\ \mathbf{A} &= \mathrm{Softmax}\left(\mathbf{K}^T\mathbf{Q}\right) \quad \mathcal{X}_{i,\mathrm{msa}} = \mathbf{V}\mathbf{A}\end{aligned} \tag{8}$$

in which θ_Q, θ_K, and θ_V represent the learnable parameters.

During the spectral degradation process, unlike low-frequency geometric features such as shape, high-frequency information representing the target's spectral characteristics undergoes severe loss and mixing, which is decomposed into the null space component by RND. Its spectral continuity and the uniqueness of characteristic bands exert a decisive influence on the finally generated HSI. Therefore, how to extract them from mixed and highly distorted unpaired data is paramount. To this end, NukesFormers incorporates Gabor kernel-based multihead self-attention (G-MSA) and nonuniform KANs networks (Nukes). These two components focus, respectively, on extracting components within target frequency intervals and mining two types of spectral properties, ultimately achieving the extraction of highly compressed null space features.

To mine deeper into high-frequency features in varying scenes, a scene-adaptive Gabor kernel is embedded by a dynamic convolution layer [41], [42], [43] to facilitate feature generation from various directions and frequency-domain intervals. Notably, the range of intervals is contingent upon the specific objects under consideration

$$g_m\left(x, y | f_m\right) = \exp\left(-\frac{x_\theta^2 + \left(y_\theta^2 / f_m\right)^2}{2\sigma^2}\right) \tag{9}$$

where f_m is the mth dynamic frequency parameter to control the interval of the Gabor kernel, $x_\theta = x\cos(\theta) + y\sin(\theta)$, and $y_\theta = -x\sin(\theta) + y\cos(\theta)$. And x and y mean the horizontal and vertical position coordinates in spatial dimension, respectively.

As depicted in Fig. 2(c), the Gabor kernels are embedded into a dynamic depthwise convolution to lightweight. Subsequently, the multiple windows captured are mapping to one learnable frequency-domain component for every channel

$$\begin{aligned}\mathbf{V}_{f,m} &= \mathbf{V} * g_m(x, y) \quad \mathbf{V}_f = \left\{\mathbf{V}_{f,1}, \mathbf{V}_{f,2}, \ldots,\right\}\\ \mathcal{X}_{i,\mathrm{fry}} &= \mathrm{Conv}\left(\mathrm{GELU}\left(\mathbf{V}_f\right)\right)\end{aligned} \tag{10}$$

by which the frequency value $\mathbf{V}_f \in \mathbb{R}^{H\times W\times CM}$ is mapping to $\mathcal{X}_{i,\mathrm{fry}} \in \mathbb{R}^{H\times W\times C}$. After that, the final feature in G-MSA is obtained with the element-wise addition operation

$$\mathcal{X}_{i,f} = \mathcal{X}_{i,\mathrm{fry}} + \mathcal{X}_{i,\mathrm{msa}}. \tag{11}$$

3) Nonuniform KAN Networks: In the conventional MetaFormer architecture, MSA generates multiple attention maps that are typically integrated via concatenated MLP or CNN-based FFN. However, they often neglect the intricate nature of spectral features and the capacity to capture high-dimensional characteristics critical for hyperspectral tasks. Therefore, UnHIG imposes heightened requirements on the capacity of building high-dimensional mapping in the FFN.

In response to the challenges of high-dimensional spectral fitting in UnHIG, we introduce a lightweight and interpretable framework, Nukes. Leveraging advanced KANs that streamline the traditionally redundant linear weights and nonlinear activation functions into a compact set of initial learnable parameters ω. These parameters are used within a recursive approach to generate multilevel B-spline basis functions $\mathbf{N}_{i,p}$

$$\mathbf{N}_{i,0}(x) = \begin{cases}1, & x_i \le x < x_{i+1}\\ 0, & \text{otherwise.}\end{cases} \tag{12}$$

$$\begin{aligned}\mathbf{N}_{i,p}(x) = &\frac{x - x_i}{x_{i+p} - x_i}\mathbf{N}_{i,p-1}(x)\\ &+ \frac{x_{i+p+1} - x}{x_{i+p+1} - x_{i+1}}\mathbf{N}_{i+1,p-1}(x).\end{aligned} \tag{13}$$

While KANs offer a powerful recursive framework, they are inherently GPU-unfriendly due to recursive dependencies. To enhance parallel computation and efficiency, we reformulate the recursive process into matrix form as (14), achieving significantly improved computational performance on GPU architectures

$$\begin{aligned}\mathbf{N}_p &= \mathbf{T}_p\mathbf{M}_{B,p}\\ \mathbf{T}_p &= \left[\mathbf{u}^p, \mathbf{u}^{p-1}, \ldots, 1\right]\end{aligned} \tag{14}$$

where $\mathbf{N}_p$, $\mathbf{T}_p$, and $\mathbf{M}_{B,p}$ represent the weight matrix of control points, multistage index matrix, and B-spline basis matrix constructed by learnable parameters, respectively.

However, capturing essential high-dimensional spectral features in UnHIG remains challenging due to limited nonlinear representation and uniform reference point placement along the coordinate axis. To address this, we propose a nonuniform control point generator (NCPG), which dynamically adjusts control points' relative attraction and exclusion forces within the Nukes curve. During training, control points $\mathbf{P}_i$ are adaptivelyconcentrated on the most salient regions of the high-dimensional spectral features. Considering that the HSI itself has the property of spectral continuity (similarity of adjacent spectra) and characteristic bands. All nodes are not uniformly distributed in the high-dimensional space; instead,

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they are adaptively and nonuniformly embedded in the high-dimensional manifold through a set of vectors. Therefore, in regions with high spectral similarity, their distribution is relatively scattered. For channel signals with key features, nodes are densely arranged to exploit the deeply compressed null space

$$\mathbf{Nuk}(x) = \frac{\sum_{i=0}^{n} \mathbf{N}_{i,p}(x) \cdot \omega_i \cdot \mathbf{P}_i}{\sum_{i=0}^{n} \mathbf{N}_{i,p}(x) \cdot \omega_i} \tag{15}$$

where $\mathbf{Nuk}(x)$ is the nonuniform B-spline curve.

Finally, the other branch with a spectral CNN extracts the low-frequency information to complement the geometric features as

$$\begin{aligned}\left[\mathcal{X}_{i,\mathrm{up}}, \mathcal{X}_{i,dw}\right] &= \mathrm{Chunk}\left(\text{Spec-Conv}\left(\mathcal{X}_{i,f}\right)\right) \\ \mathcal{X}_{i+1} &= \alpha \mathbf{Nuk}\left(\mathcal{X}_{i,f}\right) + \mathcal{X}_{i,\mathrm{up}} \cdot \mathrm{sigmoid}\left(\mathcal{X}_{i,dw}\right)\end{aligned} \tag{16}$$

where α is the learnable ratio.

Specifically, when the B-spline function fits the high-dimensional feature of HSI targets [such as inverse spectral response functions (SRFs)], feature channels often exhibit more complex nonlinear topological properties, which implies that they demand more nodes for representation. On the other hand, for channels with higher spectral similarity or continuity, their properties tend to be more locally continuous, thus requiring relatively fewer nodes. Under this prior, the model has been observed to adaptively aggregate and disperse without relying on an extra loss function, leading to a significant enhancement in its representational capability.

The generation and activation process of the NCPG is formulated as follows:

$$\mathbf{P}_i = \mathrm{Pooling}\left(\mathbf{N}_i(x)\right) \cdot \mathbf{P}_{\mathrm{ori}\ i} \tag{17}$$

in which $N(x)$, ω_i, and $P_{\mathrm{ori},i}$ represent the weight matrix of control points, ith nonlinear activation functions into a compact set of initial learnable parameters, and ith learnable vector in NCPG. This means the max-pooling function in the NCPG.

In summary, Nukes aims to minimize parameter usage while maintaining interpretability, allowing it to adaptively fit and learn critical high-dimensional features and mapping relationships in hyperspectral tasks.

C. Dual-Dimensional Contrastive Prior Module

In pursuit of stable constraint, the DCPM is employed to facilitate interaction between the two domains by building the relationship between the RGB and HSI of distinct scenes with range space. However, distinguishing from existing methods [23], [44], the DCPM introduces dual-dimensional contrastive priors, which are dedicated to resolving while concurrently handling mutual information from both domains, which are embedded with significantly different feature distributions.

Consequently, the DCPM incorporates two auxiliary encoders ($\mathcal{F}_A(\cdot)$ and $\mathcal{F}_B(\cdot)$) that map the output of the final stage's Nuk-MSA to the spectral and geometric features of latent spaces, in which the auxiliary encoders are constituted by an MLP with the activation function, illustrating with the "degradation" phase, the features extracted from both domains are divided into patches. Then, a single patch from one domain is designated as the query code $\mathbf{f}$. Subsequent patches from both domains are then partitioned into positive $\mathbf{f}^+$ and negative codes $\mathbf{f}^-$. It is noteworthy that patches from both domains are utilized in distinguishing positive from negative samples, marking a deviation from previous methods [27]. After filters are applied to randomly extract relevant patches, the resultant samples undergo spectral contrastive learning

$$\begin{aligned}&\mathcal{L}_{\mathrm{spec}}\left(\mathcal{G}_{R\to H}, \mathcal{G}_{H\to R}\right) \\ &= \mathbb{E}_{h\sim H, r\sim R}\left[-\log \frac{\mathrm{sam}\left(\mathbf{f}, \mathbf{f}^+\right)}{\mathrm{sam}\left(\mathbf{f}, \mathbf{f}^+\right) + \sum_{i=1}^{N} \mathrm{sam}\left(\mathbf{f}, \mathbf{f}_i^-\right)}\right]\end{aligned} \tag{18}$$

where the sam represents the spectral angle mapper (SAM) to quantify spectral similarity between query code and $\mathbf{f}^+$ or $\mathbf{f}^-$. Additionally, we also align the geometric distribution in the "reconstruction" phase with the relative geometric position

$$\begin{aligned}&\mathcal{L}_{\mathrm{geo}}\left(\mathcal{G}_{R\to H}, \mathcal{G}_{H\to R}\right) \\ &= \mathbb{E}_{h\sim H, r\sim R}\left[-\log \frac{\mathrm{sim}\left(\mathbf{f}, \mathbf{f}^+\right)}{\mathrm{sim}\left(\mathbf{f}, \mathbf{f}^+\right) + \sum_{i=1}^{N} \mathrm{sim}\left(\mathbf{f}, \mathbf{f}_i^-\right)}\right]\end{aligned} \tag{19}$$

where $\mathrm{sim}(u^T, v) = \exp((u^T v)/(\|u\|\|v\|\tau))$ represents the cosine similarity function and τ is the temperature parameter.

D. More Details of the Framework

Given the absence of paired RGB–HSIs, constructing pixel-wise constraints directly becomes infeasible. As a result, NukesFormers are compelled to build more implicit or indirect constraints to optimize the $\mathcal{G}_{R\to H}$ and $\mathcal{G}_{H\to R}$.

1) Adversarial Loss: Obviously, the absence of direct constraints on the intermediate outputs, $\hat{\mathbf{X}}_f$ and $\mathbf{Y}_f$, causes the reconstruction results of NukesFormers to resemble features of the latent space more than HSIs. Recognizing this, we introduce the adversarial loss. By feeding both the reconstructed and real dual-domain images from the dataset into $\mathcal{L}_{\mathrm{adv}}$, we aim to ensure that the final results are HSIs embodying the right spatial and spectral distribution, not merely feature maps

$$\begin{aligned}\mathcal{L}_{\mathrm{adv}}^{\mathrm{hsi}} &= \mathbb{E}_{rh\sim H} \log D_H(\mathbf{X}) \\ &\quad + \mathbb{E}_{fh\sim H}\left[\log\left(1 - D_H\left(\hat{\mathbf{X}}_f\right)\right)\right].\end{aligned} \tag{20}$$

Similar to other cycle-based methods [23], [45], $\mathcal{L}_{\mathrm{adv}}^{\mathrm{hsi}}$ makes the reconstructed-HSI similar to the real distribution in the HSI domain. And the dual-domain adversarial loss is calculated by $\mathcal{L}_{\mathrm{adv}} = \mathcal{L}_{\mathrm{adv}}^{\mathrm{hsi}} + \mathcal{L}_{\mathrm{adv}}^{\mathrm{rgb}}$.

2) Nondegraded Loss: Furthermore, inspired by the non-degradable properties of the spectrum, we introduced non-degraded loss to independently constrain $\mathcal{G}_{R\to H}$ and $\mathcal{G}_{H\to R}$. Specifically, when an RGB image undergoes the process represented by $\mathcal{G}_{H\to R}$, which denotes spectral degradation, its spectral features remain aligned, and the converse is also true. Consequently, $\mathcal{L}_{\mathrm{nde}}$ is used to separately ensure their functional reliability

$$\begin{aligned}\mathcal{L}_{\mathrm{nde}} &= \mathbb{E}_{r\sim R}\left[\left\|\hat{\mathcal{G}}_{R\to H}(\mathbf{X}) - \mathbf{X}\right\|_2^2\right] \\ &\quad + \mathbb{E}_{h\sim H}\left[\left\|\hat{\mathcal{G}}_{H\to R}\left(\hat{\mathbf{Y}}\right) - \hat{\mathbf{Y}}\right\|_2^2\right].\end{aligned} \tag{21}$$

Note that $\hat{\mathcal{G}}_{R\to H}$ and $\hat{\mathcal{G}}_{H\to R}$ represent that $\mathcal{G}_{R\to H}$ and $\mathcal{G}_{H\to R}$ removed the first up and downsampling operator, respectively.

To sum up, $\mathcal{L}_{\mathrm{total}}$ is defined as

$$\begin{aligned}\mathcal{L}_{\mathrm{total}} &= \lambda_1 * \mathcal{L}_{\mathrm{cyc}} + \lambda_2 * \mathcal{L}_{\mathrm{nde}} \\ &\quad + \lambda_3 * \mathcal{L}_{\mathrm{adv}} + \lambda_4 * \mathcal{L}_{\mathrm{spec}} + \lambda_5 * \mathcal{L}_{\mathrm{geo}}\end{aligned} \tag{22}$$

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where λ_i is the hyperparameters. The specific setting in our experiment is provided in the next session.

IV. Experiments

A. Experimental Setting

1) Hyperspectral Datasets: In comparison experiments, we select four benchmark datasets, consisting of NTIRE 2020 Clean Track [46], NTIRE 2022 [47], CAVE [48], and IEEE grss_dfc_2018 [49]. In the NTIRE 2020 Clean Track dataset, there are 450 paired RGB–HSIs, and we partition them into two nonoverlapping sets of 225 RGB images and 225 HSIs, respectively, which are used to construct the UnHIG experiment. We also adopt the standard validation set consisting of 10 pairs of images. Additionally, all images share the same spatial resolution of 512×482 and 31 bands spanning the 400–700-nm range. Similar to NTIRE 2020, the NTIRE 2022 dataset has the same spatial and spectral resolution as the former. A corpus of 900 pairs of images in the training set and an additional 50 pairs in the validation set is utilized in experiments. Among these, 450 scene pairs from the training set are designated as RGB images, while the remaining 450 pairs serve as the HSI components. The CAVE is a slim dataset for (HSI tasks, including 32 512×512 HSIs with 31 bands (400–700 nm), in which the 24 patches are selected randomly for training and the rest of them are prepared to validate. Similarly, it is also split into an unpaired HSI–RGB evenly as a training set. The IEEE grss_dfc_2018 is a remote sensing dataset for change detection. Its data were acquired by the ITRES CASI 1500 sensor, which has a spectral range of 380–1050 nm and 48 bands. The original data consist of 27 hyperspectral patches with a size of 512×512. We randomly selected 12 nonoverlapping patches as the HSI training data and another 12 as the MSI part, while the remaining 3 patches were used as the validation set. In addition, considering that the original data include near-infrared bands, we additionally selected the 42nd band as a supplement to the MSI, resulting in a total of four bands used as the MSI input.

2) Hyperparameters Setting: During the experiment, the optimal learning rate was adopted for all the experiments. Based on this foundation, **Ours** adopts $n = \{1, 2, 4, 2, 1\}$. Furthermore, to balance the learning process between the generator and the discriminator, the default loss weights were set as $\lambda_1 = \lambda_3 = 1.0$, $\lambda_2 = 0.5$, and $\lambda_4 = \lambda_5 = 0.25$.

Regarding the experimental environment, our work was conducted on the PyTorch platform utilizing the NVIDIA 4090 GPU. The Adam optimizer and cosine learning rate, consisting of $\beta_1 = 0.9, \beta_2 = 0.999$, and initial learning rate $lr_{\text{init}} = 0.0002$, were employed as the primary optimization mechanisms during the experiment. Additionally, our methods were trained with batch_size = 1 and epoch = 200. The specific sensitivity of hyperparameters is discussed in the ablation experiments.

3) Evaluation Criteria: In our comparative and ablation experiments, we use four evaluation criteria to thoroughly verify the effectiveness of our method, including root-mean-square error (RMSE), mean relative absolute error (MRAE), structure similarity index measure (SSIM), and peak signal-to-noise ratio (PSNR).

It is noted that the experimental results provide both the PSNR and RMSE metrics simultaneously. This practice stems from their complementary characteristics: the RMSE directly measures the average pixel-level error at the original data scale. The PSNR evaluates the relative signal-to-noise ratio, inherently decoupling from data intensity scales.

4) Comparative Methods: To demonstrate the efficiency and adaptability of our proposed framework, we integrate HSCNN+ [50], HRNet [6], HDNet [7], AWAN [20], Restormer [51], MST++ [8], and SPECAT [9] as feature extraction units using existing state-of-the-art methods, where the comparative backbones are embedded into the baseline framework of unpaired HIG. The details of the baseline are described in the ablation experiments. In addition, another unpaired method, UnGUN [14], has with strong dependency on SRF priors and a complete endmember library as default input. While UnGUN achieves parameter efficiency (notably lower than existing methods), its performance degrades significantly in practical scenes where SRF or endmember libraries are unavailable.

B. Performance Evaluation

In this section, we segment the experimentation into three distinct parts: to validate the reliability and robustness of the NukesFormers against existing models, and the ablation experiments are implemented to prove the effectiveness of the proposed modules. Finally, we set the ratio experiments to judge the flexibility of NukesFormers. The results offer a holistic testament to the efficacy of the NukesFormers under the UnHIG paradigm, underscoring its superior performance in practical applications.

1) Comparative Experiments: In this section, we evaluate NukesFormers against seven SOTA supervised HIG networks and existing UnHIG frameworks as shown in Fig. 2. The former comparison aims to validate the effectiveness of the proposed Nukes module and dual contrastive learning constraints integrated into the traditional unpaired frameworks, while the latter demonstrates that NukesFormers achieves superior performance across multiple datasets under UnHIG settings.

To ensure fairness, all supervised methods uniformly adopt identical configurations: first upsampling layer and final convolutional mapping layers are removed, retaining only their backbone architectures. They are inserted into a framework to represent the Nukes, with cycle consistency loss, nondegraded loss, adversarial loss, and spatial contrastive learning constraints. For datasets lacking essential priors needed by the existing UnHIG methods, we build the prior with machine learning for fair comparative results.

Comparative experiments were conducted on the official validation sets of three benchmark datasets: NTIRE 2020 Clean Track, NTIRE 2022 Spectral Reconstruction, and CAVE. The quantitative results presented in Table I demonstrate that our method achieves state-of-the-art performance across all evaluation metrics. Specifically, on the NTIRE 2020 Clean Track dataset, our approach outperforms the second-best method by significant margins of 16.1% in RMSE, 3.2% in MRAE, 4.3% in PSNR, and 1.8% in SSIM. Similar superior performance is observed on the NTIRE 2022 Spectral Reconstruction dataset, where we achieve improvements of 8.5% (RMSE), 44.9% (MRAE), 1.8% (PSNR), and 2.1% (SSIM)

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TABLE I

EXPERIMENTAL RESULTS OF NTIRE 2020 CLEAN, NTIRE2022, AND CAVE DATASETS IN UNHIG. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED AND UNDERLINED. THE EXISTING UNPAIRED FRAMEWORK, UNGUN, IS DIVIDED WITH THE LINE. * REPRESENTS THE SRF IS UNAVAILABLE IN THIS DATASET. ‡ INDICATES THAT ONLY THE MODEL ENSEMBLE IS USED IN THE UNPAIRED FRAMEWORK

Method	NTIRE 2020 Clean				NTIRE 2022				CAVE			
	RMSE	MRAE	PSNR	SSIM	RMSE	MRAE	PSNR	SSIM	RMSE	MRAE	PSNR	SSIM
HSCNN+‡	0.0976	0.3312	19.53	0.7002	0.1124	1.2473	20.68	0.4107	0.0739	0.9421	22.86	0.7123
HRNet‡	0.0597	0.2346	24.93	0.8329	0.0991	0.9019	22.10	0.5282	0.0690	0.7801	24.09	0.7392
HDNet‡	0.0467	0.1763	27.57	0.8862	0.0646	0.7537	25.22	0.8049	0.0402	0.7214	28.14	0.8869
AWAN‡	0.0475	0.1897	26.98	0.8711	0.0778	0.7916	23.69	0.7143	0.0528	0.7461	26.02	0.7951
Restormer‡	0.0389	0.1523	29.20	0.9082	0.0573	0.7285	26.69	0.8350	0.0399	0.7126	28.75	0.9098
SPECAT-L‡	0.0408	0.1617	28.89	0.9102	0.0637	0.7528	26.22	0.8288	0.0412	0.7154	28.30	0.9111
MST++‡	0.0384	0.1542	29.24	0.9113	0.0609	0.7002	26.46	0.8264	0.0405	0.6733	28.28	0.9107
UnGUN	0.0362	0.1675	28.63	0.9248	0.0793*	0.7832*	25.04*	0.7942*	0.0477	0.6995	27.93	0.8832
Ours	**0.0312**	**0.1476**	**30.55**	**0.9420**	**0.0528**	**0.4829**	**27.17**	**0.8526**	**0.0386**	**0.5781**	**29.57**	**0.9275**

TABLE II

EXPERIMENTAL RESULTS OF GRSS_DFC_2018 IN UNHIG. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED AND UNDERLINED. ‡ INDICATES THAT ONLY THE MODEL ENSEMBLE IS USED IN THE UNPAIRED FRAMEWORK

Model	MRAE	PSNR	SSIM	SAM
HRNet‡	0.1733	20.75	0.8429	8.97
HDNet‡	0.1064	24.29	0.8748	7.85
SPECAT-L‡	0.1190	25.35	0.8820	7.52
MST++‡	0.0927	25.14	0.8771	6.10
UnGUN	0.0990	27.35	0.8979	7.10
Ours	**0.0873**	**27.58**	**0.9010**	**5.26**

over the runner-up method. The CAVE dataset evaluation further confirms our method's advantages, with performance gains of 3.4%, 16.5%, 2.8%, and 1.8% in RMSE, MRAE, PSNR, and SSIM, respectively, compared to the second-best approach.

Comprehensive evaluations on UnHIG benchmark tasks confirm that NukesFormers achieves SOTA performance. While the UnGUN framework exhibits competitive parameter and FLOPs efficiency, its heavy reliance on supplementary prior knowledge, such as complete endmember libraries and SRF, constrains practical applicability. Notably, on datasets with accessible priors (e.g., NTIRE 2020 Clean Track and CAVE), UnGUN delivers competitive results, yet suffers significant performance degradation (RMSE: +0.0265, MRAE: +0.3003, PSNR: −2.13, SSIM: −0.0584) on datasets lacking such priors.

As evidenced by the visual error analysis in Fig. 3, the proposed NukesFormers demonstrates superior performance by achieving the lowest error in both the global and edge regions. Compared to conventional CNN-based approaches, Transformer-based methods exhibit substantial improvement in overall error reduction due to their global feature extraction capabilities, which is further corroborated by Fig. 4. However, residual noise persists in object edgy regions (e.g., vehicle edges), attributable to the absence of dedicated mechanisms for spectral channel feature processing. The UnGUN baseline achieves comparable performance to NukesFormers in scenes with sufficient prior guidance, where the comprehensive endmember library provides spectral characteristics for target features (e.g., dolls) and enables effective feature generation. Nevertheless, this approach exhibits significant degradation in practical applications due to noise amplification in detailed regions when the prior becomes incomplete.

In the grss_dfc_2018 dataset, we obtained the results of comparative experiments as shown in Table II, where Ours achieved the best performance across all four metrics. Specifically, our method, which only relies on unpaired HSIs and MSIs (including an additional near-infrared band), outperforms the existing UnHIG method UnGUN, which depends on a complete endmember library. Furthermore, compared with other HIG models, our method exhibits smaller errors in the high-frequency information of edge regions due to the scene adaptability of Nukes, as illustrated in Fig. 5. The proposed NukesFormers demonstrate superior performance compared to existing Transformer and CNN-based architectures across critical metrics including parameter efficiency, computational complexity (FLOPs), and inference speed, as quantitatively validated in Table III. This advancement stems from the strategic integration of GPU-optimized Nukes modules, which synergizes the parameter efficiency of KAN-based designs with the computational parallelism of grouped matrix operations. Furthermore, the introduction of a attention-like mechanism enables adaptive allocation of node resources, prioritizing spectral channel containing critical features while ensuring optimal utilization of limited computational parameters.

2) Ablation Analysis: We conduct comprehensive ablation studies on the NTIRE 2022 and CAVE datasets to validate the effectiveness of our proposed module, with experiments organized into three components: module ablation analysis, comparison between Nukes and traditional KANs, and hyperparameter sensitivity evaluation.

The module ablation study comprises six configurations as detailed in Table IV. NukesFormers is compared against five progressively degraded variants: 1) "DCPM-S" and "DCPM-G" eliminate the spectral and geometric encoders, respectively, from the contrastive learning framework; 2) "G-MSA" substitutes our attention mechanism with standard self-attention; 3) "Nukes" replaces our proposed components with a cascaded CNN; and 4) "Base" combines all aforementioned ablation settings. Experimental results demonstrate that our contrastive learning framework yields substantial performance gains, particularly in spectral understanding, with relative improvements

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TABLE III

RATIO EXPERIMENTAL RESULTS OF THE NTIRE 2022 DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED AND UNDERLINED. * REPRESENTS THE SRF IS UNAVAILABLE IN THIS DATASET. ‡ INDICATES THAT ONLY THE MODEL ENSEMBLE IS USED IN THE UNPAIRED FRAMEWORK

Method	Params(M)		FLOPs(G)		Inference	NTIRE 2022 1:2				NTIRE 2022 1:5			
	train	val	train	val	time(ms)	RMSE	MRAE	PSNR	SSIM	RMSE	MRAE	PSNR	SSIM
HSCNN++‡	6.09	2.71	6.42	44.62	6.96 ± 0.02	0.1476	1.5998	18.42	0.3361	0.1836	2.8293	16.08	0.4838
HRNet‡	44.41	41.16	50.75	44.92	12.67 ± 0.20	0.1164	1.1595	21.03	0.6081	0.1947	1.9815	19.45	0.4835
HDNet‡	6.15	2.67	49.46	43.75	8.56 ± 0.17	0.0816	0.7720	24.54	0.7694	0.1078	1.0371	21.84	0.5539
AWAN‡	40.48	17.512	316.36	286.39	14.14 ± 0.02	0.1039	0.9651	22.74	0.7382	0.1455	1.2228	21.11	0.5424
Restormer‡	10.00	6.79	12.27	9.94	16.19 ± 0.34	0.0759	0.7430	25.04	0.7756	0.0919	0.9481	22.07	0.6297
SPECAT-L‡	4.59	1.46	5.49	4.02	11.75 ± 0.25	0.0771	0.7858	24.69	0.7852	0.1002	0.9533	21.95	0.6020
MST++‡	6.18	1.62	7.69	6.19	13.30 ± 0.23	0.0793	0.8284	24.68	0.7691	0.0993	1.0006	21.37	0.5710
UnGUN	**0.15**	**0.15**	**1.79**	1.79	8.31 ± 0.43	0.0855*	0.8196*	23.11*	0.7052*	0.1094*	1.1743*	21.03*	0.4359*
Ours-Slim	0.38	0.16	2.89	**1.36**	**5.19 ± 0.21**	0.0713	0.6852	25.96	0.7817	**0.0786**	**0.7333**	23.87	0.7091
Ours	4.54	1.35	4.01	2.56	5.41 ± 0.11	**0.0658**	**0.6123**	**26.78**	**0.8099**	0.0794	0.7428	**23.63**	**0.7183**

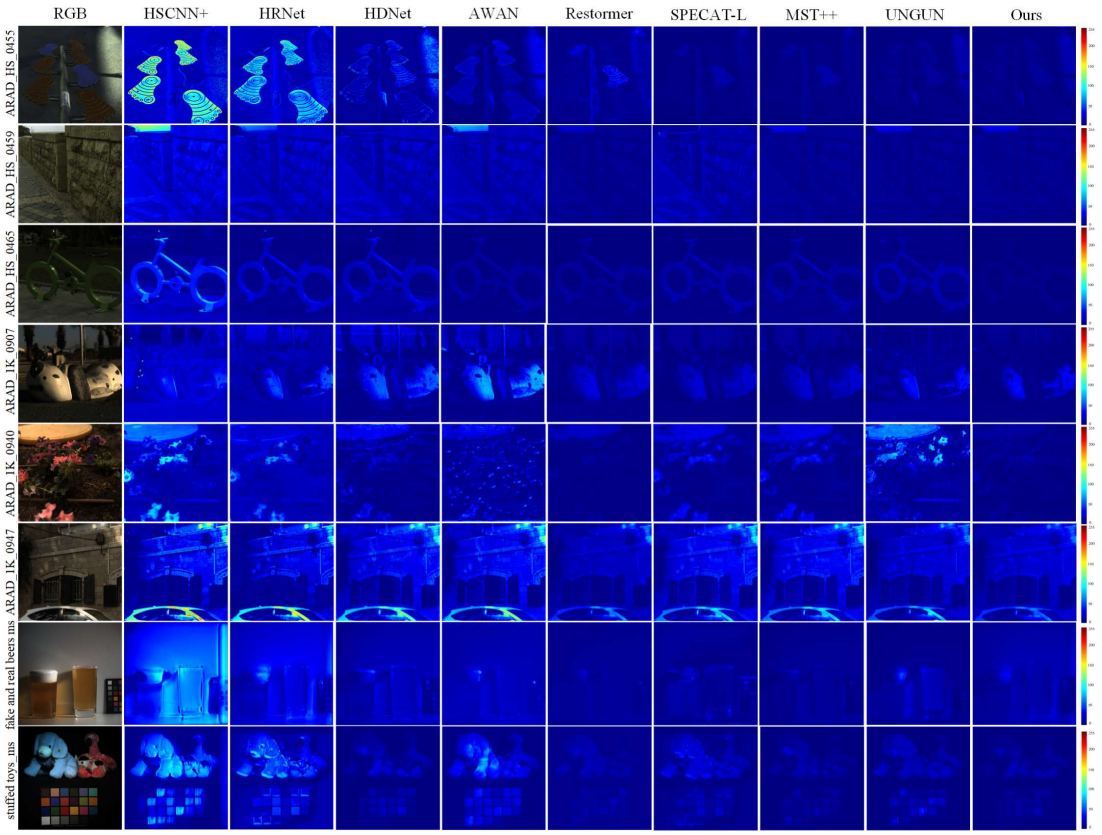


Fig. 3. *Visual Comparison Results*. Lines 1–3: the RMSE error map of the 20-h band on three distinct validation images in NTIRE 2020 Clean. Lines 4–6: the RMSE error map of the 21st band on three distinct validation images in NTIRE 2022. Lines 7 and 8: the RMSE error map of the 5th band on two distinct validation images in CAVE.

TABLE IV

ABLATION EXPERIMENTAL RESULTS OF THE NTIRE 2022 DATASET. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED AND UNDERLINED, RESPECTIVELY

Abla-tion	Params train(M)	infer time	NTIRE 2022 RMSE	MRAE	PSNR	SSIM
Base	6.01	14.68	0.0742	0.8001	25.80	0.7653
Nukes	5.87	14.76	0.0723	0.7996	25.99	0.7815
G-MSA	**4.42**	**5.35**	0.0574	0.5139	27.09	0.8334
DCPM-G	4.54	5.41	0.0634	0.5816	26.93	0.8357
DCPM-S	4.53	5.41	0.0623	0.6027	25.83	0.8174
Ours	4.54	5.41	**0.0528**	**0.4829**	**27.17**	**0.8526**

of 18.0% (RMSE), 24.8% (MRAE), 4.9% (PSNR), and 4.1% (SSIM) over "DCPM-S." The "G-MSA" exhibits degraded frequency feature extraction capability, causing performance drops of 8.7%, 6.4%, 0.3%, and 2.3% across respective metrics. "Nukes" not only increases parameter overhead but also significantly degrades reconstruction quality, and as visualized in Fig. 4, the ablated model fails to capture fine-grained foreground features, producing blurred foreground–background boundaries.

To validate the efficacy of the proposed nonuniform B-spline function and control point operator integrated into Nukes for UnHIG, we conduct ablation experiments by substituting Nukes with conventional KANs (Ours-KANs and Ours-Nukes), in Table V. As evidenced by the results in Table IV, while Nukes introduces additional parameters through the

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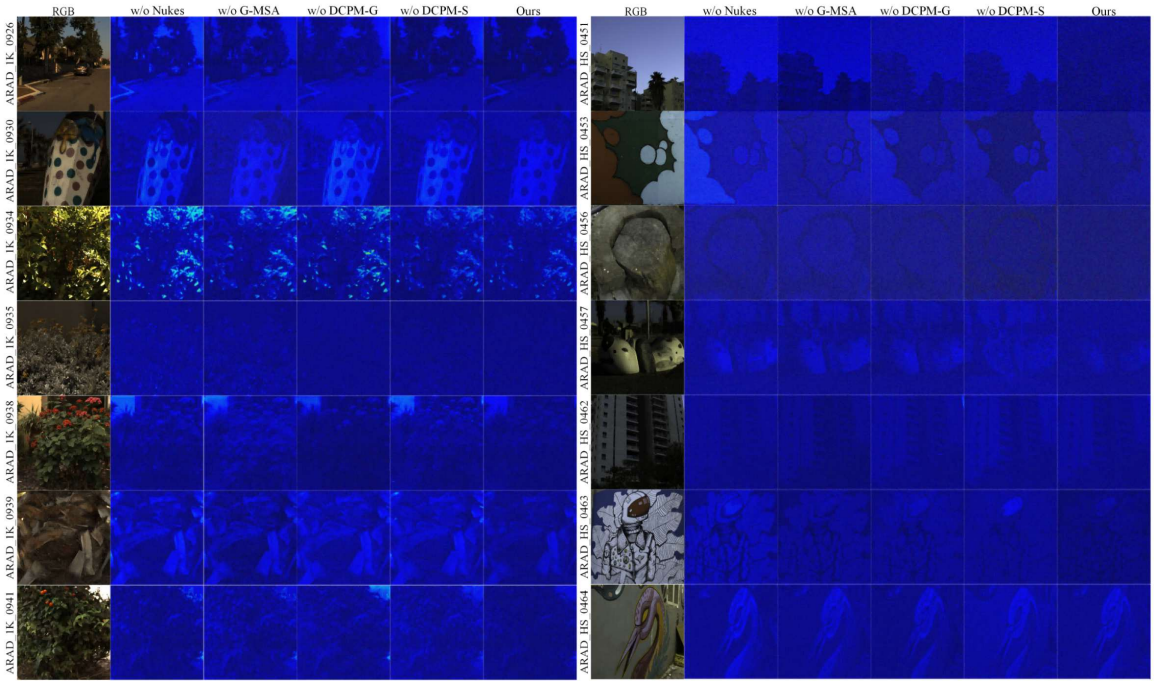


Fig. 4. *Visual Comparison Results*. Lines 1–7: the RMSE error map of seven distinct validation images in NTIRE 2022. Columns 1 and 7: the RGB images of fourteen different scenes. Columns 2–5 and 8–11: the corresponding RMSE error map lacks a key block. Columns 6 and 12: the RMSE error map of NukesFormers.

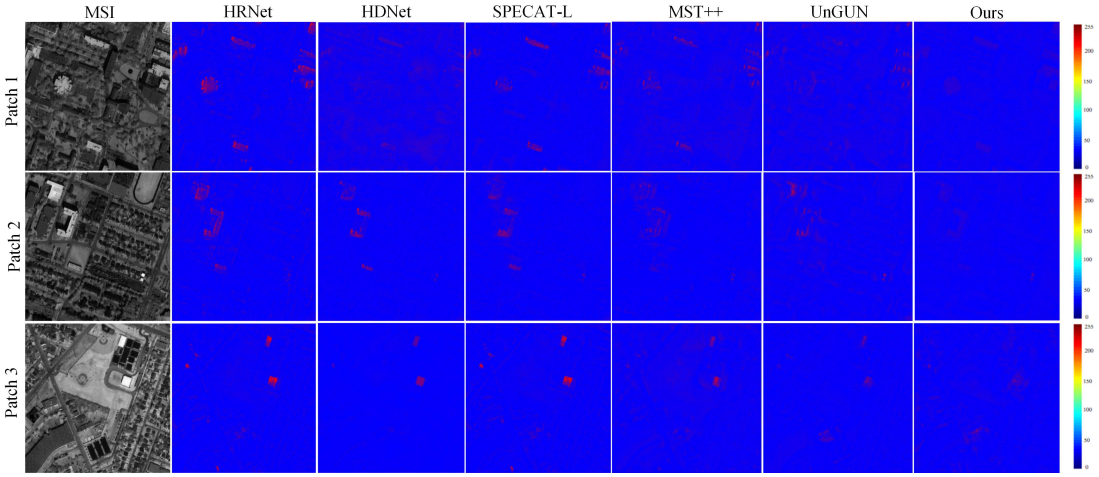


Fig. 5. *Visual Comparison Results*. The RMSE error map of the three distinct validation patches in grss_dfc_2018.

TABLE V

ABLATION EXPERIMENTAL RESULTS ABOUT THE NUKES AND KANS. THE BEST AND SECOND-BEST RESULTS ARE HIGHLIGHTED AND UNDERLINED, RESPECTIVELY

Model	Params train(M)	NTIRE 2022		CAVE	
		MRAE	PSNR	RMSE	PSNR
KANs	**2.66**	0.7882	24.07	0.0544	26.45
Nukes	<u>2.91</u>	0.7523	25.61	0.0512	27.03
Ours-KANs	4.27	<u>0.6452</u>	<u>26.65</u>	<u>0.0414</u>	<u>27.85</u>
Ours-Nukes	4.54	**0.4829**	**27.17**	**0.0386**	**29.57**

control point matrix, it delivers significant performance gains of 33.6% (RMSE) and 1.9% (PSNR) on the NTIRE 2022 dataset.

In terms of inference speed, compared with the existing MHSA, our G-MSA introduces a slight inference delay (0.06), which is attributed to the additional computational load brought by the extra Gabor Kernel embedding. For the experimental results of "w/o Nukes," it is more GPU-friendly compared with the conventional FFN. A large number of matrix and vector operations not only reduces the number of parameters but also significantly reduces the inference delay by 36.65%.

3) Ratio Analysis: To investigate the influence of HSI proportion on the UnHIG framework and assess the proposed module's adaptability to target-domain (HSI) scenes with limited data, we conducted experiments varying the ratio of HSI to RGB samples. As detailed in Table III, we partitioned the official NTIRE 2022 training set into two nonoverlapping

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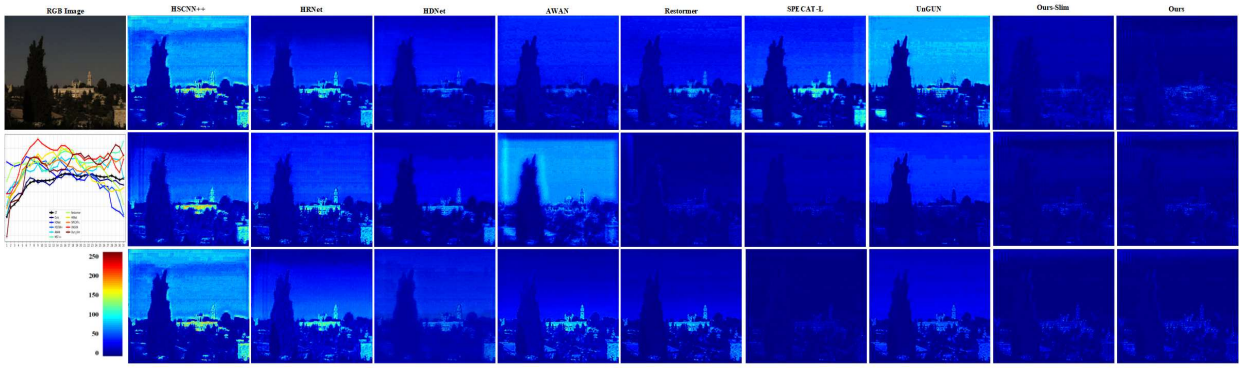


Fig. 6. Ratio experimental results on the NTIRE 2022 Hyperspectral dataset. Comparisons of the RMSE error maps in the 5th, 18th, and 25th spectral bands of the HSI (ARAD_1K_0948) with a 1:2 ratio (HSI:RGB). The spectral curves with a global average pooling are shown in the bottom left.

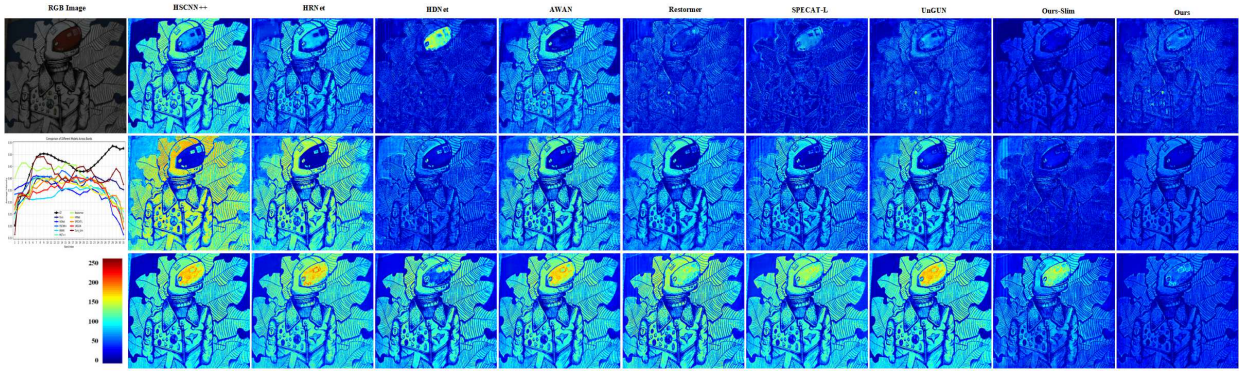


Fig. 7. Ratio experimental results on the NTIRE 2022 Hyperspectral dataset. Comparisons of the RMSE error maps in the 5th, 18th, and 25th spectral bands of the HSI (ARAD_1K_0912) with a 1:5 ratio (HSI:RGB). The spectral curves with a global average pooling are shown in the bottom left.

subsets with HSI:RGB ratios of 1:2 and 1:5, where the smaller subset represents HSI data. This progressive increase in data sparsity effectively simulates a real-world engineering application with limited hyperspectral samples.

Furthermore, we implemented a lightweight variant of NukesFormers (with rough set hyperparameter $\lambda = 1$) to examine the relationship between model complexity and HSI proportion. Experimental results demonstrate consistent performance degradation across all models under HSI feature sparsity. The conventional cycle-consistency-based approach exhibited the most significant performance drop: compared to the 1:1 baseline, its top-performing Restormer showed degradation of 32.46%, 1.99%, 6.18%, and 7.11% at a 1:2 ratio, escalating to 60.38%, 30.14%, 17.31%, and 24.59% at a 1:5 ratio.

The UnGUN framework demonstrated better robustness with degradation rates of 7.82%, 4.65%, −7.71%, and −11.21% (1:2) and 37.49%, 16.01%, and 45.12% (1:5), attributable to its compact architecture and poor performance in origin. NukesFormers also showed relatively stable performance with degradation metrics of 24.62%, 26.80%, 1.44%, and 5.01% (1:2) and 50.38%, 53.82%, 13.03%, and 15.75% (1:5).

Similarly, the visualizations, as shown in Figs. 6 and 7 demonstrate that as the proportion of HSI decreases, reconstruction errors exhibit a marked increase across all spectral channels. Error improvement is particularly pronounced in longer wavelength regions (25 channels) with diminishing HSI proportion, accompanied by significant degradation of high-frequency details in edge and foreground regions. Furthermore, models with increased parameter counts and architectural complexity show more substantial performance degradation under ratio variation. Notably, Ours-Slim achieves performance comparable to Ours at reduced HSI proportions (1:5), indicating that while larger model capacities yield stronger performance, they necessitate more diverse and information-rich target-domain features.

V. Conclusion

In this work, we introduce NukesFormers, a novel framework for UnHIG. The proposed method leverages the RND methodology to systematically partition the complex spectral distribution of objects into range space and null space. By incorporating nonuniform KANs, the framework effectively extracts and compensates for the spectral attributions within the null space. Furthermore, our model seamlessly integrates the HSI and RGB (or MSI) domains through cross-domain contrastive learning, facilitating enhanced spectral representation and domain alignment. Comparative experiments highlight the superior performance of NukesFormers, underscoring its robustness, reliability, and consistency across diverse scenes.

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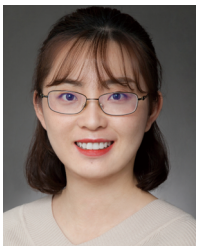
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